

2.1 Consider the following five observations. You are to do all the parts of this exercise using only a calculator.

x	y	$x - \bar{x}$	$(x - \bar{x})^2$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
-1	1	-2	4	-1	2
0	0	-1	1	-2	2
$\sum x_i = 5$	$\sum y_i = 10$	$\sum (x_i - \bar{x}) = 0$	$\sum (x_i - \bar{x})^2 = 10$	$\sum (y_i - \bar{y}) = 0$	$\sum (x_i - \bar{x})(y_i - \bar{y}) = 8$

(a) $\bar{x} = 1, \bar{y} = 2$

(b) $b_2 = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2} = \frac{8}{10} = 0.8$

$b_1 = \bar{y} - b_2 \bar{x} = 2 - 0.8 \times 1 = 1.2$

(c) $\sum_{i=1}^5 x_i^2 = 3^2 + 2^2 + 1^2 + (-1)^2 + 0^2 = 15$

$\sum_{i=1}^5 x_i y_i = 3 \times 4 + 2 \times 2 + 1 \times 3 + (-1) \times 1 + 0 \times 0 = 18$

$\sum_{i=1}^5 x_i^2 - N \bar{x}^2 = 15 - 5 \times 1^2 = 10 = \sum_{i=1}^5 (x_i - \bar{x})^2$

$\sum_{i=1}^5 x_i y_i - N \bar{x} \bar{y} = 18 - 5 \times 1 \times 2 = 8 = \sum_{i=1}^5 (x_i - \bar{x})(y_i - \bar{y})$

(d) $s_y^2 = \frac{10}{4} = \frac{5}{2}$ $s_{xy} = \frac{8}{4} = 2$

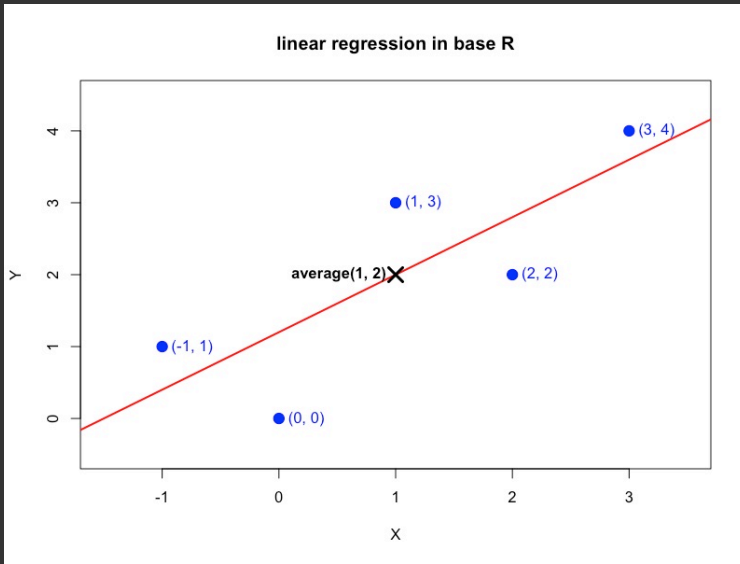
$s_x^2 = \frac{10}{4} = \frac{5}{2}$ $r_{xy} = \frac{s_{xy}}{s_x s_y} = \frac{2}{\sqrt{5/2} \sqrt{5/2}} = \frac{4}{5}$

$CV_x = 100 \left(\frac{s_x}{\bar{x}} \right) = \frac{100 \sqrt{5}}{1} = 50 \sqrt{10}$

median $(x) = 1$

x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i \hat{e}_i$
3	4	3.6	0.4	0.16	1.2
2	2	2.8	-0.8	0.64	-1.6
1	3	2	1	1	1
-1	1	0.4	0.6	0.36	-0.6
0	0	1.2	-1.2	1.44	0
$\sum x_i = 5$	$\sum y_i = 10$	$\sum \hat{y}_i = 10$	$\sum \hat{e}_i = 0$	$\sum \hat{e}_i^2 = 3.6$	$\sum x_i \hat{e}_i = 0$

(e) (f)



(g) $b_1 = 1.2$
 $b_2 = 0.8$

$\bar{x} = 1$

$\bar{y} = 2$

$\bar{y} = b_1 + b_2 \bar{x} \Rightarrow \bar{y} = 2 = b_1 + b_2 \bar{x} = 1.2 + 0.8(1) = 2$

(h) $\bar{\hat{y}} = \frac{\sum_{i=1}^5 \hat{y}_i}{N} = \frac{3.6 + 2.8 + 2 + 0.4 + 1.2}{5} = 2 = \bar{y}$

(i) $\hat{\sigma}^2 = \frac{\sum \hat{e}_i^2}{N-2} = \frac{3.6}{3} = 1.2$

(j) $\text{Var}(b_2) = \frac{\hat{\sigma}^2}{\sum (x_i - \bar{x})^2} = \frac{1.2}{10} = \frac{3}{25}$

$\text{Se}(b_2) = \sqrt{\text{Var}(b_2)} = \sqrt{\frac{3}{25}} = \frac{\sqrt{3}}{5}$